Computational Modeling of Charged Particle Identification in Detectors

Vijay Shah

July 2, 2026

Abstract

Charged particles lose energy as they traverse high-density materials in particle detectors, allowing for the identification of new particles. The Bethe-Bloch formula approximates the energy loss across distances for particle velocities $\beta > 0.05$. In this paper, we demonstrate the effects of different materials that deposit lost particle energy using a Python script, showing that lighter particles with the same initial energy as heavier particles travel farther due to having a much larger initial velocity. In addition, we demonstrate the annihilation of antimatter when it stops and demonstrate the stochastic nature of weakly interacting massive particles (WIMPs), a theorized version of dark matter. The results demonstrate how particles can be identified by energy loss and flight distance, giving a strong understanding of particle detector physics.

1 Introduction

As charged particles move through materials, their energy E decreases due to electromagnetic (EM) interactions with the electrons in the material's atoms [1]. This energy loss is important for identifying particles and tracking in modern detectors, such as the Compact Muon Solenoid (CMS) and A Toroidal LHC ApparatuS (ATLAS). Both CMS and ATLAS have four layers to their detectors: a tracking layer (designed to reveal the paths of electrically charged particles as they trigger electrical signals), an EM calorimeter (designed to stop electrons and photons), a hadron calorimeter (designed to stop hadrons such as protons), and a muon chamber (designed to track muons) [1].

Most calorimeters are designed to fully stop particles that pass through, therefore measuring their total energy. In the event that an antiparticle is stopped, an annihilation event occurs, where the antiparticle annihilates with an atom, generating a spike in the deposited energy. Calorimeters used in experiments such as A Large Ion Collider Experiment (ALICE) at CERN consist of a high-density material - such as lead - that allows particles to rapidly lose energy - along with materials such as liquid argon or scintillators to actually measure the energy loss via measurable signals such as light and electrical current. The mean energy loss per distance (also known as stopping power) of particles in high-density material is modeled by the Bethe-Bloch equation [2]:

$$-\left\langle \frac{dE}{dx} \right\rangle = \frac{4\pi n z^2}{m_e c^2 \beta^2} \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 \left[\ln\left(\frac{2m_e c^2 \beta^2}{I(1-\beta^2)} \right) - \beta^2 \right], \tag{1}$$

where m_e is the mass of an electron, c is the speed of light, $\beta = v/c$ (v is the velocity of the tracked particle), z is the charge of the tracked particle, and I is the mean excitation energy of atoms in the material. The electron density n given by

$$n = \frac{N_A Z \rho}{A M_u}, \tag{2}$$

where ρ is the density of the material, Z is its atomic number, A is its atomic mass, and M_u is the molar mass constant.

Note that Eq. 1 does not apply to electrons traveling through materials, so they require a modified treatment due to their small mass; however, it is very useful for characterizing the energy loss of heavy charged particles. Particles with more charge experience a greater energy loss proportional to z^2, and particles with a larger mass but same energy as others must move with a slower velocity (lower β), so measuring time of flight becomes essential for separating particle types.

When $\beta << 1$, Eq. 1 simplifies to

$$-\left\langle \frac{dE}{dx} \right\rangle = \frac{4\pi n z^2}{m_e v^2} \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 \left[\ln\left(\frac{2m_e v^2}{I(1-\beta^2)} \right) - \beta^2 \right]. \tag{3}$$

Inside the square brackets, we can neglect β^2, yielding

$$-\left\langle \frac{dE}{dx} \right\rangle = \frac{4\pi n z^2}{m_e v^2} \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 \left[\ln\left(\frac{2m_e v^2}{I} \right) \right]. \quad (4)$$

The above equation indicates that when a particle has low energy ($\beta << 1$), the energy loss can be modeled as v^{-2} due to the logarithmic term's slow rate of change.

If a particle has high energy $\beta \to 1$, then the logarithmic term dominates, yielding a proportion of

$$-\left\langle \frac{dE}{dx} \right\rangle \sim \ln\left(\frac{2m_e c^2 \gamma^2 \beta^2}{I} \right), \quad (5)$$

indicating a slow, unbounded rise due to relativistic effects and therefore a larger number of interactions.

Evidently, both regimes indicate that the energy loss per distance decreases with increasing β before again increasing, which produces a minimum in the stopping power (the "minimum ionizing region").

On the other hand, it has been theorized that dark matter consists of uncharged particles called "weakly interacting massive particles" (WIMPs) with a mass range of 100-1000 GeV/c^2 [3]. Therefore, dark matter does not interact with the electrons in the particle detectors, making it difficult to detect directly. Instead, dark matter particles would transfer energy to atoms through collisions - while such collisions are rare, they yield spikes in the deposited energy until the dark matter particle ceases traversal.

In this paper, the mean energy loss per distance of particles is modeled for various particle types - such as matter, antimatter, and WIMPs - demonstrating how deposited energy and flight distance can be used to identify various types of particles.

2 Methods

We created a Python simulation to model the mean energy loss of the particles. All physical constants were given in SI units. We modeled a proton slowing down in high-density materials given the following parameters:

1. Tungsten ($Z = 74, A = 184, \rho = 19250, I = 727$ eV)
2. Lead ($Z = 82, A = 207, \rho = 11348, I = 820$ eV)
3. Iron ($Z = 26, A = 56, \rho = 7874, I = 260$ eV)
4. Copper ($Z = 29, A = 63, \rho = 8960, I = 290$ eV)
5. Uranium ($Z = 92, A = 238, \rho = 19100, I = 920$ eV)

All densities are given in units of kg/m^3 and all values of I were converted to J for the simulation to remain consistent with SI units. The specific detector shown in the results below consisted of a tungsten layer with thickness of 0.15 m and a lead layer with thickness of 0.05 m. We calculated the stopping power for charged particles using Eqn. 1. However, for WIMPs, a random number generator was utilized to simulate their stochastic nature. Both methods are described in further detail below.

2.1 Charged Particles: Energy vs. Distance Traveled

To model how the energy of a charged particle varied with distance traveled, particles (a proton, muon, pion, and antiproton) with initial energy $E_i = 9 \times 10^{-12}$ J were initialized. Each step for the particles' motion was set at $dx = 10$ μm to ensure sufficient resolution with energy loss. At each step, the total energy and distance traveled were updated using the following:

$$E_{i+1} = E_i - \frac{dE}{dx}\Delta x \quad (6)$$

and

$$x_{i+1} = x_i + \Delta x. \quad (7)$$

The distance traveled through each layer was also tracked and updated according to the previous equation (being reset to 0 m after crossing the entirety of one layer). In the event that the energy loss $\frac{dE}{dx}\Delta x$ was greater than the remaining energy or $\frac{dE}{dx} < 0$, the particle's energy was set to 0 J. Finally, if the particle's remaining energy was nonzero, β was updated after applying the change in energy due to Eqn. 1.

Energy deposited by the detector was also tracked for each particle. Indeed, when the antiproton's energy reached 0 J, an annihilation event was triggered due to the interactions between the antiproton and an atom, causing a spike in the deposited energy equal to

$$E_{dep} = 2m_p c^2. \quad (8)$$

2.2 WIMPs: Energy vs Distance Traveled

Due to the fact that WIMPs have no electrical charge, Eqn. 1 cannot describe its energy loss. Instead, an interaction probability was specified for WIMPs (0.001 in the results below because of the unlikeliness of collisions between WIMPs and atoms), and a random number in the interval [0,1) was generated at each

step. If the random number was less than the probability of interaction, a random energy was generated from a uniform distribution between 0.01-0.1 MeV to determine energy loss of the WIMP. In the event that the WIMP lost all of its energy, the remaining energy was set to 0 J, similar to that of the charged particles.

3 Results and Discussion

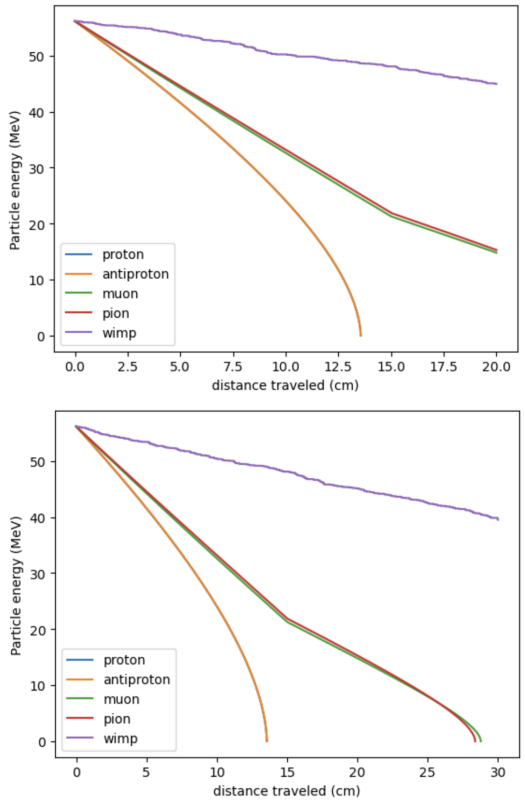


Figure 1: Graphs detailing remaining particle energy as a function of distance traveled when the detector has two layers of (a) tungsten with thickness 15 cm and lead with thickness 5 cm; and (b) tungsten with thickness 15 cm and lead with thickness 15 cm.

Fig. 1 shows graphs of the particles' remaining energy as they traversed through the detector. Though the proton and antiproton (whose curves overlap, as expected due to them having the same charge and mass) both stopped at a total distance of around 13 cm, which was still inside the tungsten layer, the muon and pion both traveled for significantly larger distances, even exiting the lead layer in Fig. 1(a). This larger distance traveled can be attributed to the muon and pion beginning with velocities far closer to the speed of light ($0.987c$ and $0.980c$, respectively) than the proton and antiproton ($0.780c$). Though Eqn. 1 indicates an unbounded, logarithmic increase in the stopping power, the significantly larger initial velocity causes them to travel through the materials faster and allows them to exit the more dense material quickly. The difference in energy for the muon and pion is small compared to their difference with the proton and antiproton due to the similar masses of the muon and pion. Nevertheless, the Bragg peak is present given sufficient detector thickness, as shown in Fig. 1(b).

In Fig. 1, lead is shown to provide a lower stopping power (represented by a flatter graph) for the muon and pion. Such results are in agreement with Eqn. 1, which describes the stopping power as proportional to $\rho\frac{Z}{A}$ of an element. The WIMP (which has a mass of 2.193×10^{-25} kg) is shown to have slow, almost steady energy loss, indicating that the energy loss is independent of the material. This was due to the interaction probability being fixed at 0.001, rather than adjusted according to the material properties.

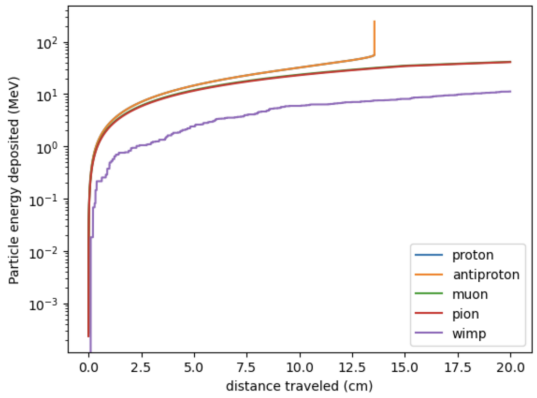


Figure 2: Graphs detailing energy deposited in the detector as a function of distance traveled when the detector has two layers of tungsten with thickness 15 cm and lead with thickness 5 cm.

Fig. 2 once again shows that the proton and antiproton stop before the muon, pion, and wimp, with the vertical axis being scaled logarithmically to better visualize the trends of energy deposition. The deposited energy of the muon and pion approach the final value for that of the proton. However, the annihilation event when the antiproton stops is represented through the spike in deposited energy, which shows how the detector could be used to identify antimatter.

4 Conclusions

In this work, we have explained the effects of high-density materials on energy loss of particles. The behavior of different particles according to the Bethe-

Bloch equation was consistent, showing that lighter particles with the same energy as a heavier particle would travel farther in a given detector. Future work includes implementing density effect corrections and extending the simulation to handle openly available data from CERN to reproduce results of particle identification.

References

[1] How a detector works. `https://www.home.cern/science/experiments/how-detector-works`.

[2] Mark Thomson. *Modern Particle Physics*. Cambridge University Press, Cambridge, United Kingdom, 2013.

[3] M. Workman et al. Review of Particle Physics: Dark Matter. *Progress of Theoretical and Experimental Physics*, 2022(8):083C01, 2022.